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ORIGINAL ARTICLE

Artificial oocyte maturation in *Patella depressa* and *Patella vulgata* using NaOH-alkalinized seawater

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Abstract
Laboratory investigations of environmental factors controlling early development and larval survival of free-spawning marine invertebrates are often limited by the difficulty of inducing spawning under laboratory conditions. Patellid gastropod species must be dissected in order to obtain the gametes from the gonads, often yielding unfertilizable immature eggs. The artificial maturation of oocytes through NaOH-alkalinization methods was investigated in order to improve laboratory experiments on larval ecology of *Patella depressa* and *Patella vulgata*. Maturation was accelerated above pH 9.0. Alkalinizing oocytes at pH 9.5 significantly enhanced *in vitro* fertilization rates, especially at higher sperm concentrations. pH 10 was harmful to eggs. Sperm dilution experiments using alkalinized oocytes suggest the use of sperm concentrations proximate to 10^6 spermatozoa ml^{-1}.

Key words: *Fertilization, larval rearing, mollusc, oocyte maturation,* Patella, *pH*

Introduction

To investigate the ecology of larval stages of benthic invertebrates in the laboratory large quantities of larvae are needed. These may be difficult or impossible to collect directly from the plankton, and the only way to isolate them is through larval rearing. In order to obtain larvae of free-spawning species, mature gametes must be obtained and fertilized. In some taxa spawning can be induced through different methods (Lillie 1915; Morse et al. 1977; Peña 1986; Kay & Emlet 2002). For most patellid limpets, however, it is necessary to dissect the gonads in order to obtain gametes (Dodd 1955; Baker 2000); this can lead to low fertilization rates (Patten 1868; Loeb 1902; Wolfsohn 1907; Smith 1935; Dodd 1955; Smaldon & Duffus 1985; Baker 2000).

Hydroxyl ions can increase the amount of fertilizable oocytes in limpets (Wolfsohn 1907). In *Patella vulgata* Linnaeus, 1758 and other prosobranchs, a pH increase can release oocytes from meiosis arrest (Guerrier et al. 1986; Catalan & Yamamoto 1993), and a natural intracellular pH rise occurs in bivalve oocytes undergoing hormone-induced maturation (Deguchi & Osanai 1995). Artificial alkalinization of oocytes has been used in shellfish aquaculture (Corpuz 1981) and cytological research (Gould et al. 2001). However, there are few methodical tests regarding the application of this technique for experimental studies in larval ecology. Smaldon & Duffus (1985) studied the effect of pH on fertilization rates in *Patella vulgata*, but they could not distinguish whether pH was acting upon the oocyte maturation process or during fertilization. Harris et al. (2002) have successfully used alkalinization to mature and fertilize oocytes of the bivalve *Pteria sterna* (Gould, 1851), and suggested its use to produce larvae for repopulation of this exploited species. Hodgson et al. (2007) have improved fertilization rates in *Patella ulyssiponensis* Gmelin, 1791 by exposing oocytes to NH_4OH, and applied this method to the study of fertilization kinetics.

We tested the effect of NaOH-alkalinization on oocyte maturation and fertilizability, for *Patella*

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depressa Pennant, 1777 and *P. vulgata*. We also determined the best pH values, immersion times and sperm concentration. To test whether embryos obtained by this method could yield normal larvae in suitable quantities for bioassays, survival of zygotes cultured in laboratory was quantified at different stages up to the late veliger stage.

Materials and methods

Effects of alkalinization on oocyte maturation

Preliminary tests. Specimens were collected intertidally in Batten Bay, Plymouth (UK) in 2003. Specimens of *Patella depressa* were collected in July, and experiments with this species were performed during August and September. Specimens of *Patella vulgata* were collected in early November, and experiments were performed during November and December. Individuals of both species were maintained in running seawater tanks in the Marine Biological Association until used for the experiments. Sperm and oocytes were obtained by dissecting the foot and pipetting the gametes from the gonads into a clean beaker. Oocytes from 3 to 5 females were rinsed and pooled together in a beaker prior to the experiments forming an oocyte suspension to be used in the experiments.

The oocyte suspension of *Patella depressa* was reduced to a minimum volume by decanting the oocytes and siphoning off the water. This suspension was then distributed equally among glass bowls containing 150 ml of pH-manipulated seawater solutions using NaOH (pH 8.5, 9.0, 9.5, 10 and 10.5) and bowls containing control seawater.

After different periods of time (0.5, 1.5, 2.5, 5.5 h), 100 oocytes from each bowl were analysed under a stereomicrospe to calculate the percentage of spherical oocytes and non-chorion-layered oocytes (these features indicate maturation of the oocyte). The experiment was stopped after 5.5 h because the cell membranes started to disintegrate at the highest pH treatments.

Following the results obtained for *P. depressa*, a similar experiment was performed in *P. vulgata* in order to test the effect of pH 9.5 for this species.

Analysis of variance. Following the preliminary results, the experiment with *Patella depressa* was repeated three times using pH 9.5 and control treatments, in order to perform a two-way ANOVA to test the effects and interaction of pH and time of immersion upon maturation. Duncan's post-hoc tests were applied. Duncan's test is a multiple range test and was performed using the software Statistica. Percentages of non-chorion-layered oocytes were arcsin-transformed to meet the analysis assumptions.

Effects of pH on fertilization rates

Preliminary tests. Glass bowls containing *Patella depressa* oocytes, submerged in NaOH-alkalinized and normal seawater, were prepared as above. After different periods of time (1, 2.5, 5.5 h), 30 ml from each glass bowl (normal seawater, pH 8.5, pH 9.5 and pH 10) were filtered through a 30 μm mesh and resuspended in 30 ml of filtered seawater. Nine millilitres of this final suspension were then transferred to three glass vials. Three males were dissected, and sperm were obtained by pipetting, using the minimum amount of seawater needed. This procedure leads to a 'dry sperm' that contains approximately 10^7 spermatozoa ml^{-1}. One millilitre from each sperm suspension was added to one vial containing the oocytes, resulting in a 10 × dilution of the 'dry sperm'. The same fertilization procedure was applied to oocytes immediately after spawning (artificially). After the experiment, actual sperm concentrations from each replica were checked by counting the number of spermatozoa using a Neubauer chamber, under a compound microscope. This was done to make sure actual sperm concentrations during the experiment were near 10^6 spermatozoa ml^{-1}.

Eight hours after sperm contact had started, fertilization rates were categorized as being Low (<10%), Moderate (>10% and <50%) or High (>50%), by observing the relative number of developing embryos.

Analysis of variance. During the experiment described above, for the 1-h treatment, 100 oocytes from each vial were analysed under a stereomicroscope and a one-way ANOVA was performed to test the effect of oocyte treatment on fertilization rates. Duncan's multiple range post-hoc tests were applied using the software Statistica.

Effects of alkalinization × sperm dilution on fertilization rates

An oocyte suspension obtained as described above was reduced to a minimum volume by decanting the oocytes and siphoning the water. Half this suspension was added to 250 ml of NaOH-alkalinized seawater at pH 9.5, and the other half was diluted in the same amount of normal seawater. After 3 h, the oocytes from both treatments were filtered and resuspended in 250 ml of normal seawater. Twenty-four small glass vials were filled with 9 ml of the alkalinized oocytes suspension, and 24 with 9 ml from the non-alkalinized oocyte suspension. Sperm solutions at seven different concentrations were prepared by taking a mixture of sperm from two males and performing 10-fold serial dilutions.

From each sperm concentration, 1 ml was added to three vials of each oocyte treatment (normal and pH-matured). One millilitre of filtered seawater was added to three egg-only control vials for each oocyte treatment, to make sure the seawater used was sperm-free. After 8 h, the percentage of developing embryos in a subsample of each replicate was measured under a stereomicroscope. Actual sperm concentrations were analysed using a Neubauer chamber under a compound microscope.

This experiment was performed both for *Patella depressa* and *Patella vulgata*. A two-way ANOVA was performed to test the effects and interactions of sperm concentration and alkalinization on fertilization rates in *P. vulgata*. In *P. depressa*, there was no fertilization in non-alkalinized oocytes at any sperm concentration, therefore a one-way ANOVA was performed for this species to test only the effect of sperm concentration upon fertilization rates in NaOH-matured oocytes. *Patella depressa* data were arcsin-transformed prior to ANOVA to meet the analysis pre-requisites. Duncan's post-hoc tests were applied for both species.

Rearing of the larvae

Oocytes from four females of *P. vulgata* were pooled together and matured artificially at pH 9.5 using NaOH-alkalinized seawater. The NaOH-matured oocytes were then fertilized in a sperm solution of 10^6 spermatozoa ml^{-1}. Approximately 4×10^4 eggs were obtained. After 5 h, sperm were washed off, and eggs were evenly distributed in three 500-ml beakers filled with seawater and left to develop at 18°C. No change of water was necessary before the trochophore stage was reached.

When the embryos reached the swimming trochophore stage, the normal swimming larvae were isolated from the abnormal ones by siphoning the upper layer of water, representing about two-thirds of the total volume from each beaker and filtering it using a mesh. All the normal larvae were resuspended and pooled together in a beaker containing 400 ml of seawater. The swimming larvae were isolated again whenever abnormal or dead larvae were detected at the bottom of the culture. When the larvae reached the veliger stage, the culture was filtered and refilled daily, and a few drops of a rich culture of *Phaoelodactylum tricornutum* was added at this point. The total number of embryos at different stages of the culture was estimated after homogenizing the culture and analysing a subsample (5 ml) under a compound microscope and a stereomicroscope microscope.

Results

Effects of alkalinization on oocyte maturation

Preliminary data. Higher pH values improved the maturation process when compared to lower pH values in *Patella depressa*. A stepwise change in the number of spherical oocytes occurred at pH 9.5, between lower pH treatments and high pH treatments (Figure 1a), but it was not possible to detect differences between the effect of pH 9.5, 10.0 and 10.5 upon shape (Figure 1b).

After 0.5 h, in lower pH treatments (control, 8.5 and 9.0), ~5% of oocytes observed were spherical, while in higher pH treatments (9.5, 10, 10.5) this value ranged between 35 and 49%. This difference was maintained during the monitoring period. After 1.5 h, at higher pH, the percentage of spherical oocytes was always greater than 70%. In contrast, at lower pH, the percentage of spherical oocytes was never greater than 30% (Figure 1a).

By analysing chorion disintegration in *P. depressa* (Figure 1b), only pH 9.5 appeared to enhance maturation. After 0.5 h, in all treatments, all oocytes still had an intact chorion. At pH 9.5, 79% of the oocytes had lost the chorion after 2.5 h, whereas in all other treatments the number of oocytes without chorion remained close to zero. After 5.5 h, chorion disintegration at pH 10 had reached the same level as at pH 9.5.

The effect of pH 9.5 upon oocyte shape in *Patella vulgata* seemed very similar and consistent over the period studied. However, there was no chorion disintegration at pH 9.5 (Figure 1c, d).

Alkalinization at pH 9.5. The experiment in *Patella depressa* showed significant effects of alkalinization (NaOH, pH 9.5) (ms = 1766 $F = 136.34$, $p < 0.001$) and immersion time (ms = 2578, $F = 19.9$, $p < 0.001$) on the rate of spherical oocytes, but no significant interaction (ms = 368, $F = 2.8$, $p = 0.051$). At all time treatments, post-hoc tests revealed a percentage of spherical oocytes significantly higher at pH 9.5 than in the control (Figure 2a). Significant effects of pH (ms = 18.640, $F = 30.29$, $p < 0.001$) and time of immersion (ms = 8.2, $F = 13.40$, $p < 0.001$) upon the percentages of oocytes without chorion, and a highly significant interaction (ms = 2.4, $F = 3.9$, $p < 0.001$), were found. Post-hoc tests revealed a significantly higher number of oocytes without chorion when they are submerged in NaOH alkalinization for 1.5 h or more (Figure 2b).

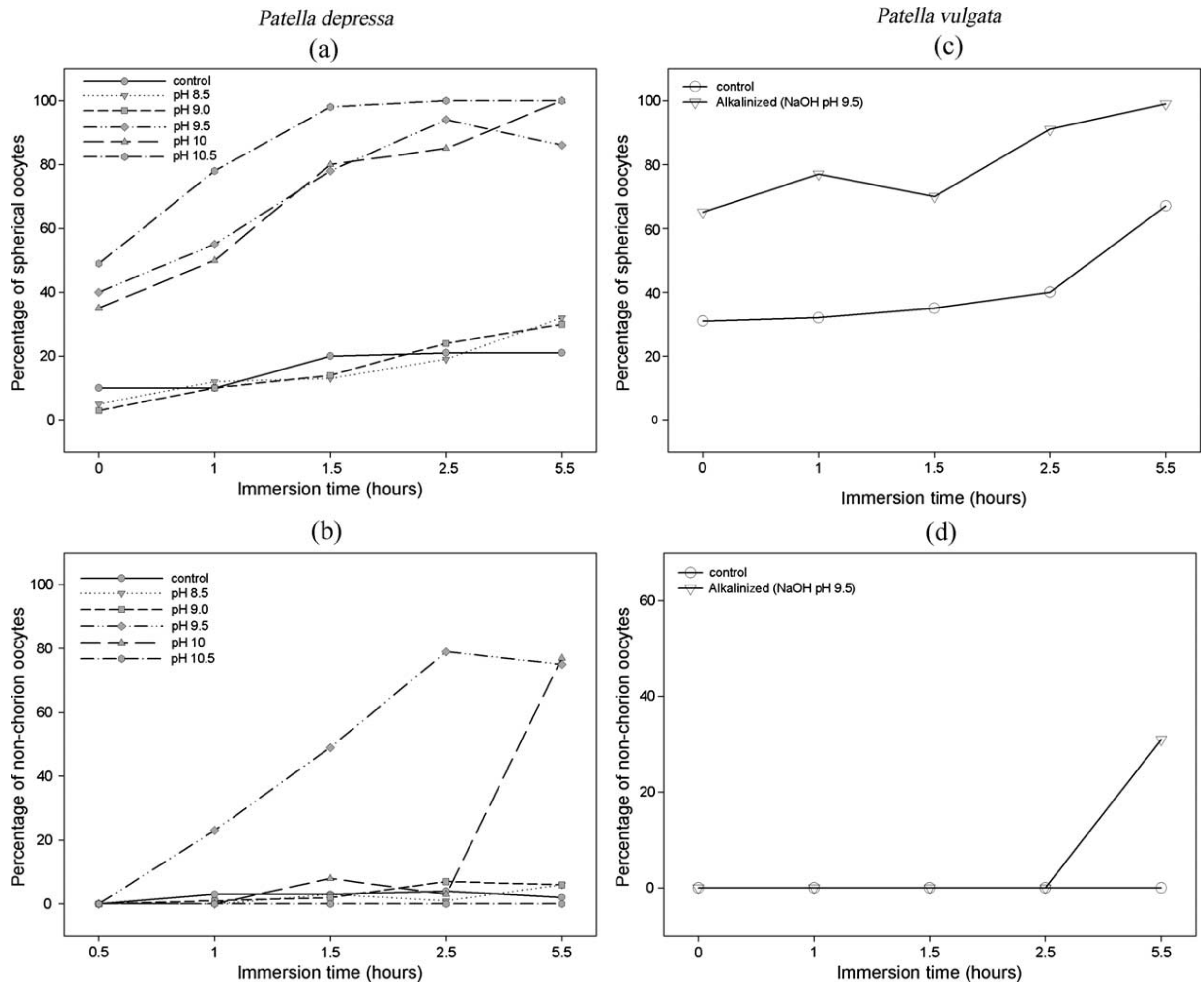

Figure 1. Pilot experiments ($n = 1$) conducted with oocytes obtained by dissection. Analysis of oocyte maturation indicators in NaOH-treated oocytes and control oocytes. (a) Percentage of spherical oocytes, with differing immersion times and seawater pHs, in *Patella depressa*. (b) Percentage of spherical oocytes, with differing immersion times in control and alkalinized (pH 9.5) oocytes, in *Patella vulgata*. (c) Percentage of non-chorion oocytes, with differing immersion times and seawater pHs in *Patella depressa*. (d) Percentage of non-chorion oocytes, with differing immersion times in control and alkalinized (pH 9.5) oocytes, in *Patella vulgata*.

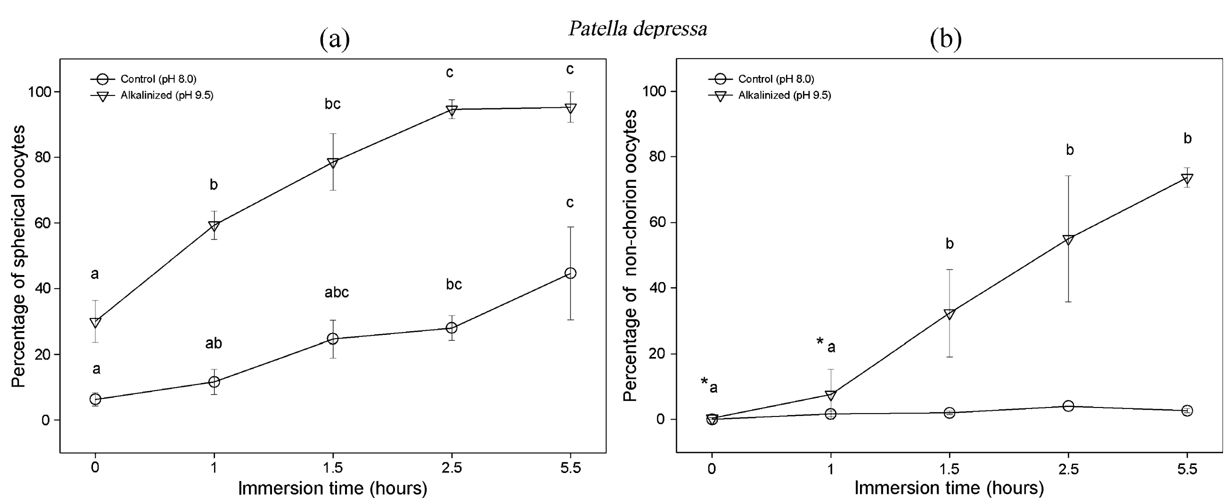

Figure 2. Maturation indicators in *Patella depressa* NaOH-treated (pH 9.5) oocytes and control oocytes. Oocytes were obtained by dissection. (a) Percentage of spherical oocytes, with differing immersion times. (b) Percentage of non-chorion oocytes, with differing immersion times. Bars represent standard errors ($n = 3$). Same letters means no significant (Duncan's test, $p < 0.05$) statistical difference within an oocyte treatment. *No significant statistical difference between oocyte treatments in a given time treatment.

Effects of alkalinization on fertilization rates

Preliminary data. Oocytes of *Patella depressa* NaOH-alkalinized at pH 9.5 yielded the best fertilization rates. Prolonged immersion periods in alkaline solutions did not lead to higher rates of fertilization. Exposure to pH higher than 9.5 reduced fertilization rates. There was no variation between the different males used within a treatment, and therefore only one symbol had to be used for each treatment in Table I.

One-hour immersion time. Samples from the 1-h treatment were counted and subjected to ANOVA and the percentages of fertilized oocytes by each male sperm were used as replicates (Figure 3). pH had a significant effect (ms = 2163, F = 494, $p < 0.001$). Post-hoc tests revealed a significantly ($p < 0.01$) higher fertilization success in oocytes alkalinized at pH 9.5 than at any other pH tested, and in oocytes alkalinized at pH 10 than at pH 9.0 or control oocytes.

Effects of alkalinization × sperm dilution on fertilization rates

The optimum sperm concentration in *Patella depressa* and *Patella vulgata* was $\sim 6 \times 10^5$ spermatozoa ml^{-1}; as the sperm diluted, fertilization success decreased rapidly (Figure 4a, b).

For alkalinized oocytes in *P. vulgata*, there was a steep rise in the fertilization rates up to a concentration of 3×10^5 spermatozoa ml^{-1}; and from there on, fertilization continued to increase, but at a much slower rate (Figure 4a). The fertilization success was on average 32% at 3×10^4 spermatozoa ml^{-1}; at 3×10^5 spermatozoa ml^{-1} it showed a rapid rise to 58%; and at 3×10^6 spermatozoa ml^{-1}, only a small rise to 66% was observed. Alkalinization had a highly significant effect (ms = 1837, F = 74.06, $p < 0.001$) on fertilization success. Post-hoc tests revealed significant differences at all concentrations from 3×10^4 sperm ml^{-1} upwards (Figure 4b). Non-alkalinized oocytes of *Patella depressa* were tested, but no fertilization occurred after three attempts.

Rearing of the larvae

Ten hours after fertilization all embryos had reached at least the 32-cell stage. After 24 h all trochophores had reached the swimming stage. Pre-torsional veligers were obtained 48 h after fertilization. Operculums were visible at 72 h, and eyes at 96 h. From the 4×10^4 early trochophores, obtained from 4 females, approximately 8×10^3 became normal late trochophores. Approximately 50% of those developed into normal veligers and underwent torsion. The culture terminated at 140 h after fertilization, when a few larvae had become sedentary, crawling on the bottom of the beaker. Thus the length of planktonic life at temperatures similar to 18°C was approximately 6 days.

Discussion

The effect of pH on maturation

Many previous studies have demonstrated that alkalinization can induce oocyte maturation or improve fertilization rates (Wolfsohn 1907; Guerrier et al. 1986; Catalan & Yamamoto 1993; Vilain et al. 1993; Harris et al. 2002; Hodgson et al. 2007). Through the analysis of external cellular characteristics and fertilization rates, we have (1) quantified how much NaOH-alkalinized seawater can improve artificial maturation of stripped-spawned oocytes obtained by dissection in two species of *Patella*, and (2) determined the optimum pH to be used to obtain high maturation and fertilization rates. This has practical implications for laboratory experiments on the environmental control of reproductive and larval success in these species.

The results suggest the most effective pH is 9.5. Lower pH is likely to have no effect on maturation and fertilization rates. On the other hand, care must be taken not to alkalinize seawater too much because pH 10 partially prevented oocytes from being fertilized or from undergoing cleavage if fertilization was successful. Although pH 10 or above seemed to be effective in inducing oocyte shape change, it did not seem to induce chorion disintegration. Our observations indicate that pH 10 may improve

Table I. *Patella depressa*. Fertilization rates of recently shed oocytes with no pre-immersion (control 1), oocytes immersed into untreated water (pH 8.0, control 2) and into NaOH-alkalinized seawater mediums of different pHs (8.5, 9.0, 9.5 and 10). Symbols: L, <10%; M, 10%–50%; H, >50%. Three replicates were used for each treatment, because there was no difference between replicates within a treatment, only one letter is shown for each treatment.

Immersion period	Control 1	Control 2 pH = 8.0	pH = 8.5	pH = 9.5	pH = 10.0
No immersion	L	–	–	–	–
1 h	–	L	L	H	M
2.5 h	–	L	L	H	L
5.5 h	–	L	L	H	L

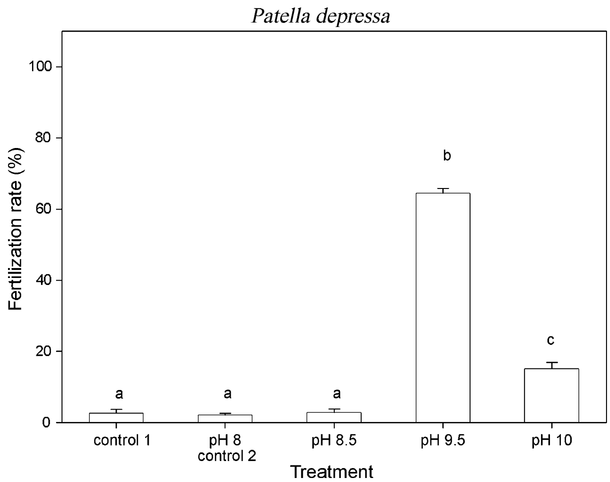

Figure 3. Mean fertilization rate of oocytes recently shed with no pre-immersion (control 1), immersed into normal seawater (pH 8.0) for 1 h (control 2), and immersed in NaOH-alkalinized seawater for 1 h at various pHs (8.5, 9.5 and 10). Bars represent standard error ($n = 3$). Different letters indicate significant differences (Duncan's test, $p < 0.05$). This test was performed using samples from the *Patella depressa* experiment described in Table I.

fertilization rates if the oocytes are not submerged to the alkalinization medium for more than 1 h, but even then the results were not as complete as in pH 9.5. These results appear to be contrary to the recommendation by Smaldon & Duffus (1985) that 'seawater must be carefully maintained at 7.9, the normal seawater pH', in their study on the effect of pH on maturation and fertilization of gametes in *Patella vulgata*. In their experiments, eggs were left for 1 h in different flasks under different pH in which later fertilization occurred by adding a few drops of sperm. Regular adjustments of pH were made during fertilization and embryonic development. Therefore it is not possible to state whether the negative results observed were a consequence of pH action on (1) the oocyte maturation process, (2) sperm activity, (3) fertilization process, or (4) cleavage stages. In our experiments, after maturation, oocytes were rinsed and fertilization always occurred under normal pH conditions. Therefore it has been possible to account for the effect of pH on maturation only, thus we conclude (1) that the detrimental effects of high pH reported by Smaldon & Duffus (1985) were a result of pH action on a stage other than oocyte maturation, and that (2) in our work improved fertilization rates were a consequence of the maturation process of oocytes triggered by a pH rise prior to fertilization.

Hodgson et al. (2006) used pH 9.0 to induce maturation in *Patella ulyssiponensis* and *Patella vulgata*, but did not test different pH, also they used NH_4OH and not NaOH. These authors quantified the improvement in fertilization rates in *Patella ulyssiponensis*, but not in *Patella vulgata*. They found fertilization rates in *P. ulyssiponensis* to improve from around 30% to around 85%. These results achieved for *P. ulyssiponensis* are similar to those reported here for *P. vulgata*. On the other hand, fertilization rates in *Patella depressa* oocytes were improved from near zero to around 60%, following NaOH-alkalinization treatment, which only worked at pH 9.5.

The pH value suggested in the present paper is similar to that used by Corpuz (1981) to induce oocyte maturation in the patellid *Cellana exarata* in the laboratory. However, the author does not say whether he tested other pH levels. Harris et al.

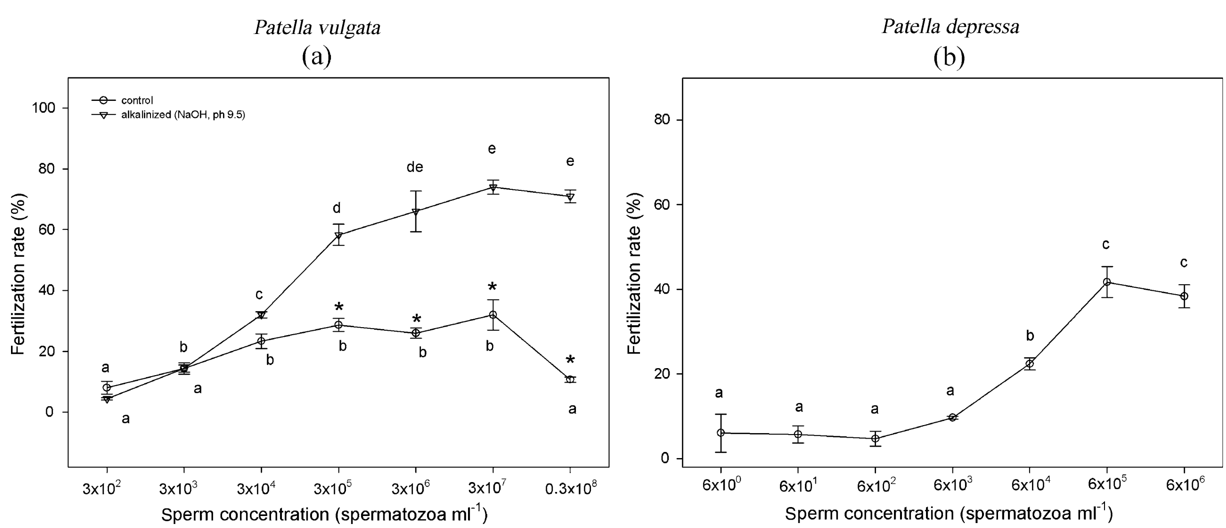

Figure 4. Effects of NaOH-alkalinization (pH 9.5) on fertilization rates of artificially spawned oocytes, at different sperm concentrations. (a) Mean fertilization rate at different sperm concentration, in alkalinized (pH 9.5) and non-alkalinized oocytes, in *Patella vulgata*. (b) Mean fertilization rate at different sperm concentration in alkalinized oocytes, in *Patella depressa*. Bars represent the standard error ($n = 3$). Significant difference between alkalinized and non-alkalinized treatments (Duncan's test, $p < 0.05$) is designated by *. Same letters indicate no significant differences (Duncan's test, $p > 0.05$).

(2002) found pH 8.0 and 8.5 to be effective in promoting Germinal Vesicle Breakdown (GVBD) in the bivalve *Pteria sterna*, but they did not test at higher pH, and like Hodgson et al. (2007) they used methods different from the one presented here to alkalinize seawater.

One previous study analysed the effect of a wide range of pH values on the maturation of oocytes of a mollusc species (Catalan & Yamamoto 1993). They analysed the effect of pH on the percentage of oocytes undergoing GVBD in *Cellana nigrolineata*. That study found the optimum pH was between 8.5 and 9.5 while higher pH had very detrimental consequences, a pattern similar to the results presented here for *Patella depressa* and *P. vulgata*. *C. nigrolineata* is similar to *P. vulgata* in terms of meiosis arrests and stage at which fertilization occurs (Catalan & Yamamoto 1993). The similarities regarding the optimum pH value for artificial maturation process might reflect a similar mechanism of meiosis reactivation in nature, in which the exact value of intracellular pH may be very important. But discussing the chemical mechanism of meiosis reactivation is beyond the scope of this paper (see Guerrier et al. 1986; Catalan & Yamamoto 1993; Vilain et al. 1993).

Optimum sperm concentrations and larval rearing

The optimum sperm concentration for fertilization success in *Patella vulgata* was first estimated by Baker (2000). She found the best fertilization success at a sperm concentration of 4.4×10^6 spermatozoa ml^{-1}, followed by a decrease in fertilization success at higher sperm concentrations. Our experiments suggest a rapid increase in fertilization success up to 3×10^5 spermatozoa ml^{-1}, followed by minor improvements in both alkalinized and non-alkalinized oocytes. As observed by Baker (2000) a rapid decrease in fertilization success in concentrations $>3 \times 10^7$ spermatozoa ml^{-1} was observed in the present study for non-alkalinized oocytes. This did not happen however in alkalinized oocytes. Both in our study, as in Hodgson et al. (2007), fertilization remained at its highest up to $>10^8$ spermatozoa ml^{-1}.

Our work suggests that the response of *P. depressa* alkalinized oocytes is very similar to that of *P. vulgata*, with a rapid increase in fertilization success from sperm concentrations similar to 10^4 spermatozoa ml^{-1} up to sperm concentrations similar to 10^6 spermatozoa ml^{-1}, followed by smaller increments in fertilization success. Although we have not quantified the proportion of abnormally developing embryos, it is well known that abnormality rate increases with sperm concentration as a result of polyspermy (Levitan 1995). To avoid or to minimize polyspermy, we recommend a sperm concentration not higher than 10^6 spermatozoa ml^{-1}, since above this value no major improvement in fertilization rates were observed in any of the species tested.

Besides successful fertilization, research on larval biology and ecology require successful development to obtain a large number of larvae in order to be able to perform bioassays. Following the steps described above it was possible to obtain 4000 veligers undergoing normal development from only four females of *Patella vulgata*. 'D' larvae of the bivalve *Pteria sterna* have been reared from alkalinized oocytes (Harris et al. 2002), but no quantification of survival during the rearing stages is presented by the authors. Corpuz (1981) observed 68% survival from trochophore to veliger stages in cultures of the patellid *Cellana exarata* using naturally matured eggs. Our result of 50% survival during the same phase is therefore satisfactory.

Therefore, by using NaOH-alkalinized oocytes it is possible to obtain a large number of trochophores and veligers, making the study of larval ecology in laboratory feasible for these species.

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